

Neural networks, machine learning and randomness

Stochastic methods for machine learning

Optional course for master students

Martin Holeňa



What will it be about?

- ◆ Estimation of latent variables from observations
- ◆ Monte Carlo Markov chain (MCMC) estimation method
- ◆ Several particular algorithms for MCMC estimation
- ◆ Efficient estimation by means of variational inference (VI)
- ◆ Evidence lower bound in VI computations
- ◆ VI with mean-field approximations and generative models

Observable and latent variables

- ◆ A situation frequent in machine learning: a long vector $X = (X_1, \dots, X_n)$ of *observable* variables is related to a vector $Z = (Z_1, \dots, Z_m)$ of *latent* variables

Recalling the Bayesian approach

- ◆ Latent variables Z help govern the distribution of data
 - they relate to the observations through $P(X|Z)$, called *likelihood*
- ◆ The dissimilarity of a distribution π_2 to a distribution π_1 is most commonly measured with the relative entropy, aka *Kullback-Leibler divergence*, from π_2 to π_1 :

$$\text{KL}(\pi_1 \parallel \pi_2) = \int \log\left(\frac{\pi_1}{\pi_2}\right) d\pi_1 = - \int \log\left(\frac{\pi_2}{\pi_1}\right) d\pi_1$$

Observable and latent variables

- ◆ A situation frequent in machine learning: a long vector $X = (X_1, \dots, X_n)$ of *observable* variables is related to a vector $Z = (Z_1, \dots, Z_m)$ of *latent* variables
- ◆ Distributions: assumed prior $P(Z)$, observed likelihood $P(X|Z)$
 - determine joint through the Bayes theorem: $P(Z, X) = P(X|Z)P(Z)$
- ◆ We are interested in obtaining the *posterior* $P(Z|X)$

Monte Carlo estimation of $P(Z|X)$

- ◆ Monte Carlo Markov chain (MCMC) method – constructs an *ergodic Markov chain* with the equilibrium distribution $P(Z|X)$

Principle of MCMC methods

- ◆ *Monte Carlo* methods: all methods based on *sampling*
 - *MCMC* methods: sampling is from a *Markov chain*
- ◆ Most commonly employed is the Metropolis-Hastings algorithm
- ◆ Standard MCMC sweeps over all data every iteration
 - disadvantage if the dataset is large

Metropolis-Hastings (MH) algorithm

- ◆ Generates a *sequence* $Z_0, Z_1, \dots, Z_m$ on a set Ω ,
the distribution of which *converges* with m
in probability to a desired *distribution* π on Ω
 - MCMC: π is the Markov chain equilibrium distribution
- ◆ Requires: m , Z_0 , an unnormalized version $\tilde{\pi}$ of the density of π ,
the density q of a distribution on Ω^2 – proposal distribution

Pseudocode of the MH algorithm

1. $k \leftarrow k + 1$
2. $Z' \sim q(\cdot | Z_{k-1})$
3. $p = \min\left(1, \frac{\tilde{\pi}(Z') q(Z_{k-1} | Z')}{\tilde{\pi}(Z_{k-1}) q(Z' | Z_{k-1})}\right)$
4. $\beta = \text{Bernoulli}(p)$
5. if $\beta = 1$ then $Z_k = Z'$
6. else $Z_k = Z_{k-1}$

Convergence of the MH algorithm

- ◆ The convergence of the distribution of $\omega_0, \omega_1, \dots, \omega_N$ to π is a consequence of the central limit theorem
- ◆ For suitable functions h , there exists $\sigma_h > 0$ such that

$$\sqrt{N} \left(\frac{1}{N} \sum_{k=0}^N h(Z_k) - \int h(z) \pi(z) dz \right)$$

has asymptotically the normal distribution $N(0, \sigma_h^2)$

MH algorithm in a Bayesian context

- ◆ For the likelihoods $(P(X_i|Z))_{i=1}^n$, define the average likelihood

$$\ell(Z) = \ell(Z)_{X_1, \dots, X_n} = \frac{1}{n} \sum_{i=1}^n \log P(X_i|Z) = \frac{1}{n} \log \prod_{i=1}^n P(X_i|Z) = \frac{1}{n} \log P(X|Z)$$

- ◆ Due to $\pi(Z) = P(Z|X) = \frac{P(X,Z)}{P(X)}$, we can choose

$$\tilde{\pi}(Z) = P(X, Z) = P(Z)P(X|Z) = P(Z)\exp(n\ell(Z))$$

- ◆ Then the 3. step of the MH algorithm reads:

$$p = \min \left(1, \frac{P(Z')}{P(Z_{k-1})} \frac{q(Z_{k-1}|Z')}{q(Z'|Z_{k-1})} \exp \left(n(\ell(Z') - \ell(Z_{k-1})) \right) \right)$$

Monte Carlo estimation of $P(Z|X)$

- ◆ Monte Carlo Markov chain (MCMC) method – constructs an *ergodic Markov chain* with the ergodic distribution $P(Z|X)$
- ◆ Draw initial $Z^{[0]} \sim q(X)$ and perform transitions $Z^{[t]} \sim q(Z^{[t]} | Z^{[t-1]}, X)$
- ◆ Can produce arbitrarily precise estimate of $P(Z|X)$
- ◆ But only after many iterations, and their number for a given precision is actually unknown

Monte Carlo Tree Search (MCTS)

- ◆ Monte Carlo method for Markov decision processes (MDPs)
 - originally in games, later planning, scheduling, combinatorial optimization
- ◆ MDPs model environments with sequential decisions to make to receive rewards after certain sequences of actions

Markov decision process

- ◆ MDP is represented as the tuple (S, A_s, P_a, R_a)
 - S – set of possible *states*, a.k.a. state space
 - $A(s)$ – set of *actions* available in $s \in S$
 - $P_a(s, s')$ – *transition* probability that the action a entails $s \rightarrow s'$
 - $R_a(s)$ – immediate *reward* for reaching state s by action a
- ◆ Deterministic case: $P_a(s, s') = 1 \Rightarrow$ in state s , leads to s'

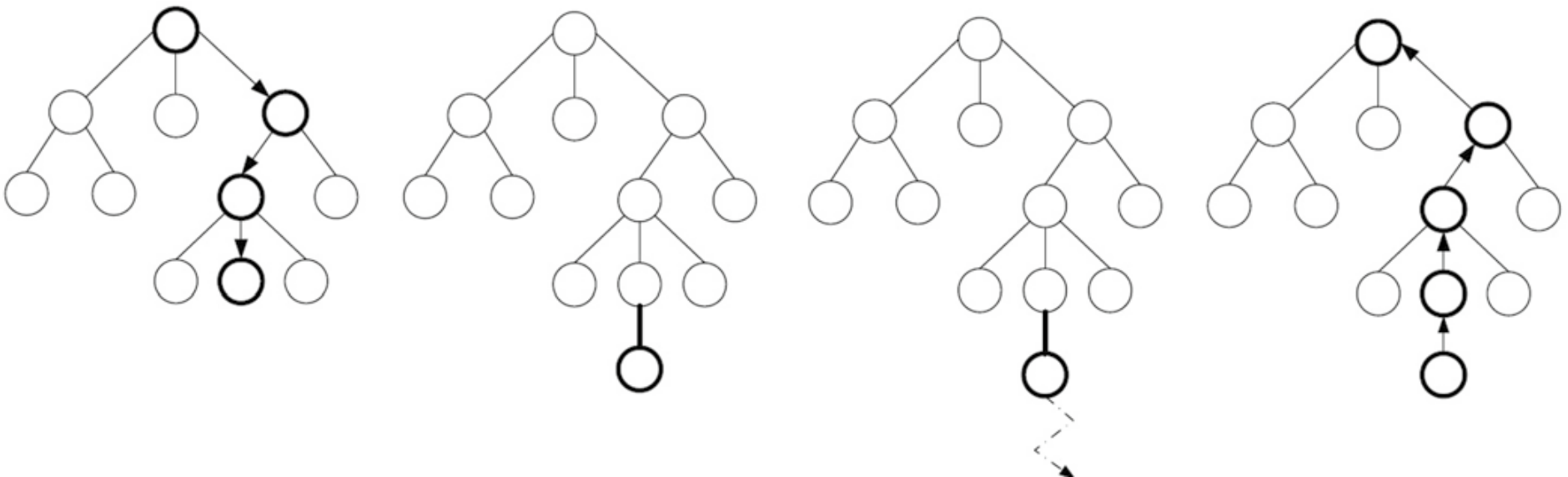
Monte Carlo Tree Search (MCTS)

- ◆ Monte Carlo method for Markov decision processes (MDPs)
 - originally in games, later planning, scheduling, combinatorial optimization
- ◆ MDPs model environments with sequential decisions to make to receive rewards after certain sequences of actions
- ◆ MCTS can be stopped anytime and *returns* $a^* = \arg \max_{a \in A(s)} Q(s, a)$
 - $Q(s, a)$ – *average reward* for action a in state s

Phases of an MCTS iteration

1. *Selection* – searching the tree portion already in the memory, from the root to a current leaf
2. *Expansion* – adding ≥ 1 children below the leaf selected before
3. *Simulation* of the problem – the MC ($\Leftarrow$ random) phase
4. *Backpropagation* – propagates the scores back along the path from the last visited leaf to the root

Run continuously in the allotted time



Selection policy

- ◆ Purpose of selection policy: balance *exploration* and *exploitation*
of insufficiently tested actions of best already found

- ◆ Most common – upper confidence bound for trees (*UCT*):

first, all actions are checked, then actions a^* fulfilling:

$$a^* = \arg \max_{a \in A(s)} \left\{ Q(s, a) + c \sqrt{\frac{\ln N(s)}{N(s, a)}} \right\}, c > 0$$

- $N(s)$ = # of state s visits, $N(s, a)$ = # of a samplings in s

Alternatives to UCT

- ◆ Upper confidence bound 1-tuned: uses $Q(s, a)$ variance σ_a

$$a^* = \arg \max_{a \in A(s)} \left\{ Q(s, a) + C \sqrt{\frac{\ln N(s)}{N(s, a)}} \min \left(\frac{1}{4}, \sigma_a + \frac{2 \ln N(s)}{N(s, a)} \right) \right\}$$

- ◆ Rapid value action estimation (RAVE): uses an additional score $Q_{\text{RAVE}}(s, a)$, updated for each action in phase 3

$$a^* = \arg \max_{a \in A(s)} \left\{ \sqrt{\frac{k}{3N(s) + k}} Q_{\text{RAVE}}(s, a) + \left(1 - \sqrt{\frac{k}{3N(s) + k}} \right) Q(s, a) \right\}$$

- k – a constant called RAVE equivalence constant

Variational inference (VI)

- ◆ A popular *alternative to MCMC* estimation
 - can not produce arbitrarily precise estimates
 - but computes them much faster, especially for complicated $P(Z|X)$ and / or for large datasets
- ◆ Variational inference is *optimization-based*
 - whereas MCMC is based on simulation

Optimization in variational inference

- ◆ Variational inference consists in solving the optimization problem

$$Q^*(Z|X) = \arg \min_{Q(Z|X) \in \mathcal{Q}} \text{KL}(Q(Z|X) \parallel P(Z|X))$$

- $\mathcal{Q}$: set of *easily computable* distributions, chosen in advance

- ◆ The minimized relative entropy can be reformulated

$$\begin{aligned} \text{KL}(Q(Z|X) \parallel P(Z|X)) &= \mathbb{E}_{Q(Z|X)} \log Q(Z|X) - \mathbb{E}_{Q(Z|X)} \log P(Z|X) = \\ &= \mathbb{E}_{Q(Z|X)} \log Q(Z|X) - \mathbb{E}_{Q(Z|X)} \log P(Z, X) + \log P(X) \end{aligned}$$

Evidence lower bound

- ◆ For observable X , call $P(X)$ its *evidence*, and define

$$\text{ELBO}(Q(Z|X)) = \mathbb{E}_{Q(Z|X)} (\log P(Z, X) - \log Q(Z|X))$$

- ◆ This definition has the following consequence:

$$\log P(X) = \text{KL}(Q(Z|X) \parallel P(Z|X)) + \text{ELBO}(Q(Z|X)) \Rightarrow$$

1. Minimal $\text{KL}(Q(Z|X) \parallel P(Z|X))$ corresponds to maximal ELBO
2. $\log P(X) \geq \text{ELBO}(Q(Z|X))$, with equality only if $Q(Z|X) = P(Z|X)$

Mean-field approximation

- ◆ The set $\mathcal{Q}$ contains *only* distributions of $Z = (Z_1, \dots, Z_m)$ that are, conditioned on X , *mutually independent*: $Q(Z|X) = \prod_{j=1}^m Q_j(Z_j|X)$
- ◆ Each latent variable Z_j has its *own distribution* $Q_j(Z_j|X)$ chosen according to the character of Z_j
 - Z_j continuous $\Rightarrow Q_j(Z_j|X)$ is often chosen Gaussian
 - Z_j categorical $\Rightarrow Q_j(Z_j|X)$ is categorical $(p_1, \dots, p_m)$, e.g., $p_j = \frac{1}{m}$

Example: Gaussian mixtures

- ◆ Consider observable variables $X_1, \dots, X_n$ with distributions that are mixtures of K univariate unit-variance Gaussian distributions

$$X_i \sim N(\mu_{c_i}, 1), i = 1, \dots, n, c_i \in \{1, \dots, K\}, K \in \mathbb{N}$$

- $\mu = (\mu_1, \dots, \mu_K)$ are the means of the K Gaussians, $\mu_k \sim N(0, \sigma^2)$
- $c = (c_1, \dots, c_n)$ are categorical random variables, $c_i \sim \left(\frac{1}{K}, \dots, \frac{1}{K}\right)$

- ◆ As latent variables for $X = (X_1, \dots, X_n)$ can serve $Z = (\mu, c)$

Properties of Gaussian mixtures

- ◆ *Joint distribution* of $(Z, X) = (\mu, c, X)$ is $P(\mu) \prod_{i=1}^n P(c_i) P(X_i | c_i, \mu)$
- ◆ The *evidence* includes computing K^n likelihoods $p(X_i | c_i, \mu)$

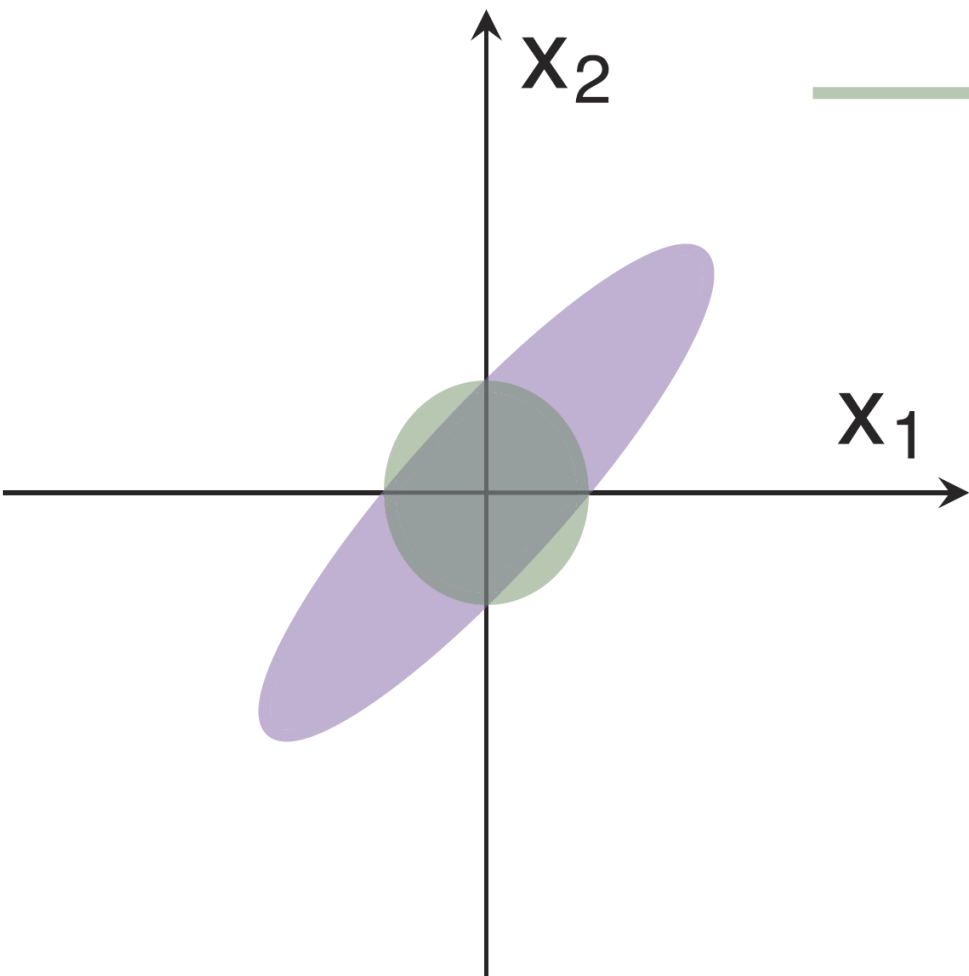
$$P(X) = \int P(\mu) \prod_{i=1}^n \sum_{c_i} P(c_i) P(X_i | c_i, \mu) d\mu = \sum_c P(c) \int P(\mu) \prod_{i=1}^n P(X_i | c_i, \mu) d\mu$$

- ◆ *Mean-field* approximation parametrized with $(m_k, s_k^2)_{k=1}^K, (\varphi_i)_{i=1}^n$:

$$Q(\mu, c | X) = \prod_{k=1}^K Q(\mu_k; m_k, s_k^2) \prod_{i=1}^n Q(c_i; \varphi_i)$$

- the distribution $Q(\mu_k; m_k, s_k^2)$ is chosen Gaussian, $N(m_k, s_k^2)$

— Exact Posterior
— Mean-field Approximation



ELBO for Gaussian mixtures

$$\text{ELBO}(Q(Z|X)) = \text{ELBO}(m, s^2, \varphi), m = (m_1, \dots, m_K), s^2 = (s_1^2, \dots, s_K^2), \varphi = (\varphi_1, \dots, \varphi_n)$$

◆ Because $Q(Z|X)$ depends on the hyperparameters m, s^2 and φ

$$\begin{aligned} \text{ELBO}(m, s^2, \varphi) &= \mathbb{E}_{Q(Z|X)} \log P(Z, X) - \mathbb{E}_{Q(Z|X)} \log Q(Z|X) = \\ &= \sum_{k=1}^K \mathbb{E}_{Q(Z|X)} [\log P(\mu_k); m_k, s_k^2] + \sum_{i=1}^n \mathbb{E}_{Q(Z|X)} ([P(X_i|c_i, \mu); m, s^2, \varphi_i] + [\log P(c_i); \varphi_i]) \\ &\quad - \sum_{k=1}^K \mathbb{E}_{Q(Z|X)} [\log Q(\mu_k); m_k, s_k^2] - \sum_{i=1}^n (\mathbb{E}_{Q(Z|X)} [\log Q(c_i); \varphi_i]) \end{aligned}$$

Combining VI with MCMC

- ◆ Markov chain variational inference (*MCVI*): every $Q \in \mathcal{Q}$ is a Markov chain $(Z_0, Z_1, \dots, Z_T)$, fulfilling $Z_T = Z$ and

$$Q(Z_0, \dots, Z_{T-1} | X, Z_T) = Q(Z_0 | X) \prod_{t=1}^T Q(Z_t | X, Z_{t-1})$$

- ◆ This can be generalized, replacing $Q(Z_t | X, Z_{t-1})$ with a specific $Q_t(Z_t | X, Z_{t-1})$ for $t = 1, \dots, T$, forming a Markov chain

$$Q(Z_0, \dots, Z_{T-1} | X, Z_T) = Q(Z_0 | X) \prod_{t=1}^T Q_t(Z_t | X, Z_{t-1})$$

Computing MCVI

- ◆ Forward distributions $Q_t(Z_t|X, Z_{t-1})$ are approximated with backward distributions $R_t(Z_{t-1}|X, Z_t)$, forming a Markov chain

$$R(Z_0, \dots, Z_{T-1}|X, Z_T) = \prod_{t=1}^T R_t(Z_{t-1}|X, Z_t)$$

ELBO extension for MCVI

- ◆ ELBO($Q(Z|X)$) has the following lower bound:

$$\text{ELBO}(Q(Z|X)) \geq \text{ELBO}(Q(Z|X)) - \text{KL}(Q(Z_0, \dots, Z_{T-1}|X, Z_T) \| R(Z_0, \dots, Z_{T-1}|X, Z_T))$$

- ◆ Combination with the original $\log P(X) \geq \text{ELBO}(Q(Z|X))$ yields:

$$\begin{aligned} \log P(X) &\geq \text{ELBO}_{\text{MCVI}} = \mathbb{E}_{Q(Z|X)} (\log P(Z, X) - \log Q(Z|X)) - \\ &\quad - \mathbb{E}_{Q(Z_0, \dots, Z_{T-1}|X, Z_T)} \left(\log Q(Z_0|X) \prod_{t=1}^T Q_t(Z_t|X, Z_{t-1}) - \log \prod_{t=1}^T R_t(Z_{t-1}|X, Z_t) \right) = \\ &\quad = \mathbb{E}_{Q(Z_0, \dots, Z_{T-1}|X, Z_T)} \left(\log \frac{P(Z, X)}{Q(Z_0|X)} + \sum_{t=1}^T \log \frac{R_t(Z_{t-1}|X, Z_t)}{Q_t(Z_t|X, Z_{t-1})} \right) \end{aligned}$$

ELBO_{MCVI} estimation algorithm

- ◆ Input: x , number of iterations T , distributions P, Q, Q_t + inverse R_t
- ◆ Initialize: sample $z_0 \sim Q(Z_0|X)$ and put $L_0 = \log P(z_0, x) - \log Q(z_0|x)$
- ◆ Iterate from $t = 1$ to $t = T$, performing:
 1. *Sample* the transition from $t - 1$ to t : $z_t \sim Q_t(Z_t|X, Z_{t-1})$
 2. *Update* the lower bound $L_t = L_{t-1} + \log \frac{P(z_t, x) R_t(z_{t-1}|x, z_t)}{P(z_{t-1}, x) Q_t(z_t|x, z_{t-1})}$
- ◆ Output: L_T — a T -iteration unbiased estimate of ELBO_{MCVI}

Computing MCVI

- ◆ Forward distributions $Q_t(Z_t|X, Z_{t-1})$ are approximated with backward distributions $R_t(Z_{t-1}|X, Z_t)$, forming a Markov chain

$$R(Z_0, \dots, Z_{T-1}|X, Z_T) = \prod_{t=1}^T R_t(Z_{t-1}|X, Z_t)$$

- ◆ If θ denotes the parameters of $Q(Z_0, \dots, Z_{T-1}|X, Z_T), R(Z_0, \dots, Z_{T-1}|X, Z_T)$, then computing MCVI consists in finding the *maximum* of $\text{ELBO}_{\text{MCVI}}$ w.r.t. θ , typically by means of $\nabla_{\theta} \text{ELBO}_{\text{MCVI}}$

VI with generative models

- ◆ Consider $P(Z, X) = P(Z)P_{\theta}(X|Z)$, where $P_{\theta}(X|Z)$ is parametrized by θ and governs a *generative model*
- ◆ Then $Q^*(Z|X)$ is sometimes called *recognition model*
 - typically, it is computed by a MLP
 - assumes that $Q \ni Q^*(Z|X)$ are normal distributions

Deep Kalman filter

- ◆ Consider the d -dimensional observed variables $(X_1, \dots, X_T)$, s -dimensional latent $(Z_1, \dots, Z_T)$, c -dimensional actions $(U_1, \dots, U_{T-1})$
- ◆ Deep Kalman filter: $(X = (X_1, \dots, X_T), Z = (Z_1, \dots, Z_T), U = (U_1, \dots, U_{T-1}))$ if

$$Z_1 \sim N(0, I_d), Z_t \sim N\left(G_\alpha(Z_{t-1}, U_{t-1}, \Delta_t), S_\beta(Z_{t-1}, U_{t-1}, \Delta_t)\right), X_t \sim \mathcal{D}\left(F_\gamma(Z_t)\right)$$

- $\mathcal{D}$: probability distribution, α, β, γ : parameters, Δ_t : duration from $t-1$ to t
- $G_\alpha, S_\beta, F_\gamma$ are usually implemented by *neural networks*

Relation to a classical Kalman filter

◆ A classical Kalman filter is governed by 2 equations:

1. $Z_t = G_t Z_{t-1} + B_t U_{t-1} + \epsilon_t$ with $\epsilon_t \sim N(0, \Sigma_t)$, a state-transition matrix

$G_t \in \mathbb{R}^{s \times s}$, and a control input model $B_t \in \mathbb{R}^{s \times c}$

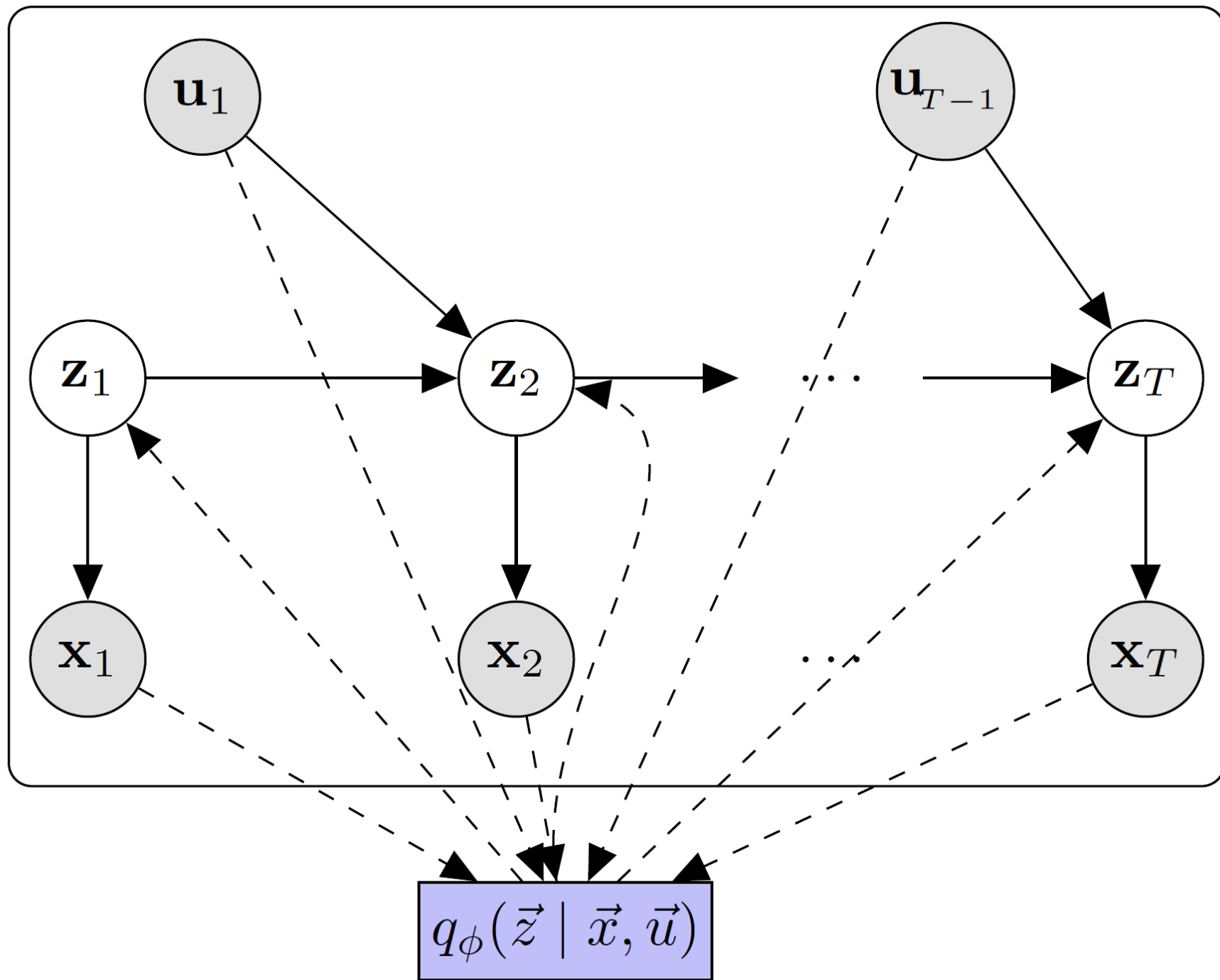
2. $X_t = F_t Z_t + \eta_t$ with $\eta_t \sim N(0, \Gamma_t)$, and an observation matrix $F_t \in \mathbb{R}^{d \times s}$

◆ This is actually a *deep* Kalman filter with

$$G_\alpha = G_t Z_{t-1} + B_t U_{t-1}, S_\beta = \Sigma_t, F_\gamma = F_t Z_t$$

VI with deep Kalman filter

- ◆ Consider a VI $q_\phi(\vec{z}|\vec{x}, \vec{u}) = Q_\phi(Z|X, U)$ with a parameter ϕ approximating a deep Kalman filter posterior $P_\theta(Z|X, U)$



VI with deep Kalman filter

- ◆ Consider a VI $q_\phi(\vec{z}|\vec{x}, \vec{u}) = Q_\phi(Z|X, U)$ with a parameter ϕ approximating a deep Kalman filter posterior $P_\theta(Z|X, U)$
- ◆ Moreover, consider only $Q_\phi(Z|X, U)$ that can be *factorized* with respect to the sequence $Z = (Z_1, \dots, Z_T)$ as follows:

$$Q_\phi(Z|X, U) = Q_\phi(Z_1|X, U) \prod_{t=2}^T Q_\phi(Z_{t-1}|Z_t, X, U)$$

Lower bound for $\log P_\theta(X|u)$

◆ For a *general* $Q_\phi(Z|X, U)$, not necessarily factorized

$$\begin{aligned}\log P_\theta(X|U) &= \log \int P_\theta(X, Z|U) dz = \log \int P_\theta(X|Z, U) P(Z|U) dz = \\ &= \log \int Q_\phi(Z|X, U) \frac{P_\theta(X|Z, U) P(Z|U)}{Q_\phi(Z|X, U)} dz \geq \\ &\geq \int Q_\phi(Z|X, U) \log \frac{P_\theta(X|Z, U) P(Z|U)}{Q_\phi(Z|X, U)} dz = \\ &= \mathbb{E}_{Q_\phi(Z|X, U)} \log P_\theta(X|Z, U) - \text{KL} \left(Q_\phi(Z|X, U) \parallel P(Z|U) \right)\end{aligned}$$

Case of factorized $Q_\phi(Z|X, u)$

- ◆ Assume the following *conditional independence* between X_t :

$$P_\theta(X|Z, U) = \prod_{t=1}^T P_\theta(X_t|Z, U) = P_\theta(X_1|Z_1) \prod_{t=2}^T P_\theta(X_t|Z_t, U_{t-1})$$

- ◆ Then the lower band *1. term* turns to $\mathbb{E}_{Q_\phi(Z|X, U)} \log P_\theta(X|Z, U) =$

$$= \mathbb{E}_{Q_\phi(Z|X, U)} \log P_\theta(X_1|Z_1) + \sum_{t=2}^T P_\theta(X_1|Z_1) \log P_\theta(X_t|Z_t, U_{t-1})$$

- ◆ And the *2. term* turns to $\text{KL}(Q_\phi(Z|X, U) \parallel P(Z|U)) =$

$$= \text{KL}(Q_\phi(Z_1|X, U) \parallel P(Z_1)) + \sum_{t=2}^T \mathbb{E}_{Q_\phi(Z_{t-1}|X, U)} \text{KL}(Q_\phi(Z_t|Z_{t-1}, X, U) \parallel P(Z_t|Z_{t-1}, Z_{t-1}))$$